



Graphically solving two linear graphs that intersect

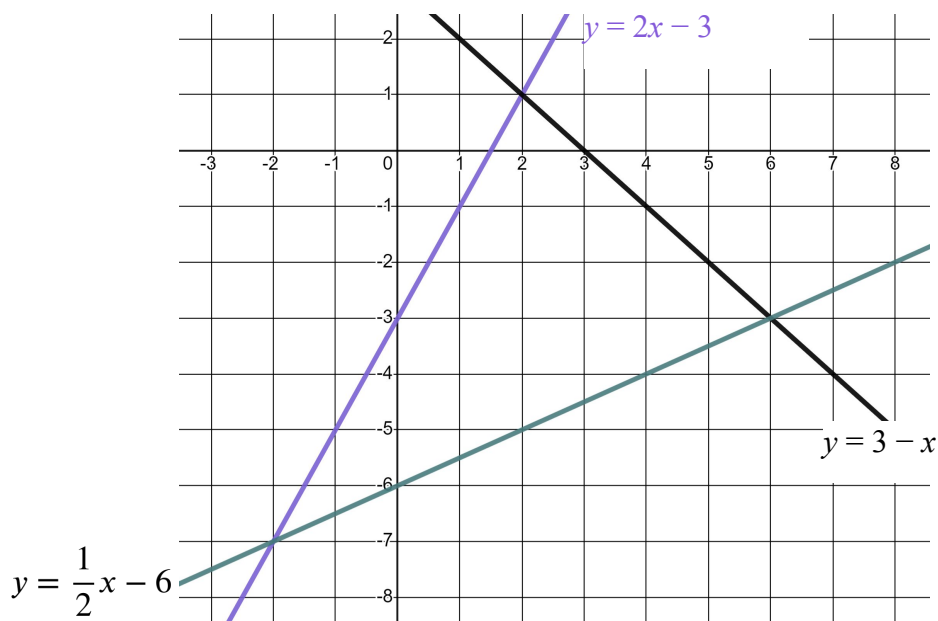
Task A: Points of intersection

1) Use these graphs to solve the equations.

a) $2x - 3 = 3 - x$

b) $2x - 3 = \frac{1}{2}x - 6$

c) $3 - x = \frac{1}{2}x - 6$



2) Use the graphs to solve these pairs of simultaneous equations.

a) $y = 2x - 3$

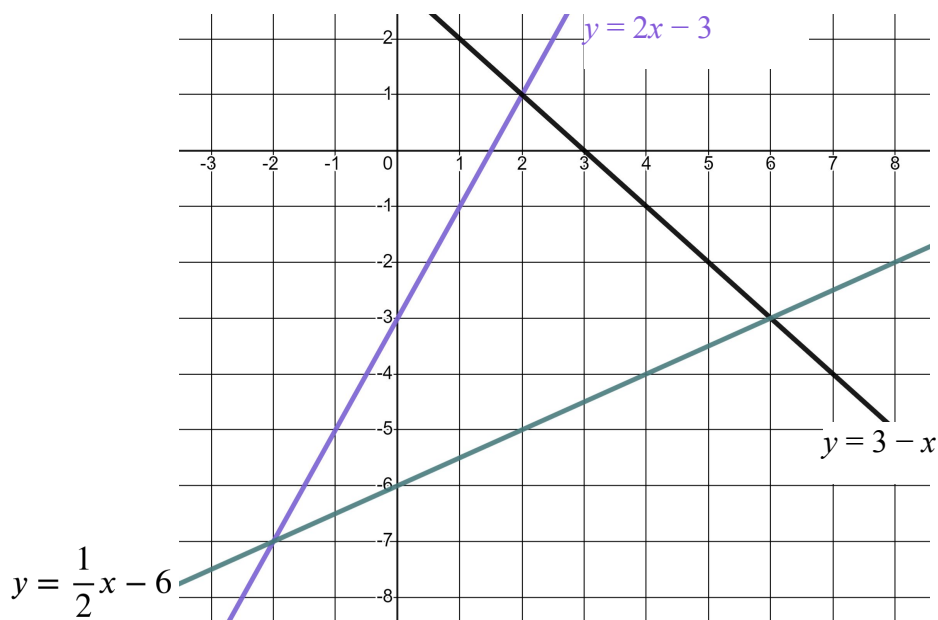
$y = 3 - x$

b) $y = 2x - 3$

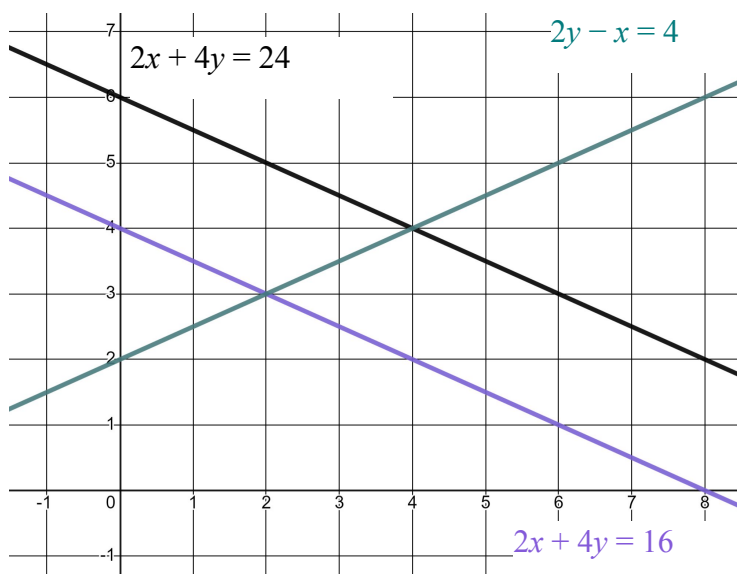
$y = \frac{1}{2}x - 6$

c) $y = 3 - x$

$y = \frac{1}{2}x - 6$



3) a) Use the graph to identify two pairs of simultaneous equations which can be solved and find the solutions.



b) Explain why a third different pair of equations can not be found and solved using these particular lines.

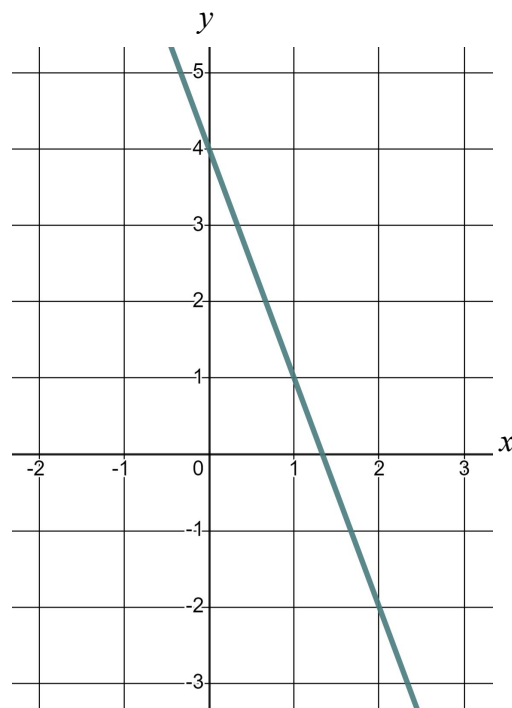
Task B: Regions of a graph

1) Here is the graph of $y = 4 - 3x$

a) Mark and label three points where $y = 4 - 3x$

b) Mark and label three points where $y < 4 - 3x$

c) Mark and label three points where $y > 4 - 3x$



2) The lines $y = 4 - 3x$ and $y = x - 4$

a) Mark and label three points where $y = 4 - 3x$ and $y > x - 4$

b) Mark three coordinates where $y < 4 - 3x$ and $y = x - 4$

c) Mark and label the one point where $y = 4 - 3x$ and $y = x - 4$

